Ricardo (Zhicong) Xie

Product / Hardware Engineering Intern | CAD, Embedded C and Mechatronics

Auckland, New Zealand | +64 29 027 29390 | zxie588@aucklanduni.ac.nz | LinkedIn: https://www.linkedin.com/in/ricardo-xie-6394643b4 | Portfolio: https://zxie588-alt.github.io | GitHub: https://github.com/zxie588-alt

Valid NZ student visa | No sponsorship required | 25h/week during semester | Full-time 17 Nov 2026 - late Feb 2027

PROFILE

- Mechanical engineering master's student with CAD, enclosure design, embedded C and sensor-system evidence.
- Relevant work: IoT sensor module, registered STM32 weighing firmware, MVHR hardware layout and workshop training.
- Project screenshots and source links: https://zxie588-alt.github.io

EDUCATION

University of Auckland - Master of Engineering Studies, Mechanical Engineering | 2026 - Present

Current study and project work in building services, smart manufacturing, CFD/thermal modelling, composite structures and engineering methods.
Relevant coursework: Advanced Industry 4.0 Smart Manufacturing, Manufacturing and Industrial Systems, Engineering Methods

Huazhong Agricultural University - Bachelor of Engineering, Mechanical Engineering / Mechatronics | 2021 - 2025

Mechanical design, CAD, electrotechnics, sensors, embedded control and mechatronic system integration.

PRACTICAL AND TEAM EXPERIENCE

- Manufacturing workshop training: prepared AutoCAD/DXF laser-cut profiles and compared drawings with cut parts.
- Engineering practice: interpreted force-displacement tests, controlled a robotic arm and led a small automotive-structure walkthrough.
- Technical communication: former student news lead with photography and project-documentation experience.

PROJECT WORK

Ruggedized Outdoor IoT Sensor Module Design Package

SolidWorks, enclosure design, DFM notes, Keil C, validation workflow

- Designed a SolidWorks enclosure with gasket stack-up, screw bosses, PCB envelope and service access.
- Prepared DFM notes for CNC review, 3D printing and assembly tolerance checks.
- Screened corner-load, strap-lug and screw-boss regions with an FEA-style check.
- Built Keil C firmware structure with sensor, battery and state-machine modules.

Dynamic Livestock Weighing System - Software Copyright Registered

STM32, HX711, embedded C, Bluetooth/serial communication

- Developed STM32/HX711 weighing firmware with real-time data transmission to an upper computer.
- Received Computer Software Copyright Registration No. 2024SR0141888 for the system.
- Supported dynamic weighing logic for improved signal stability under field-like conditions.

Small Residential MVHR / ERV Revit MEP Coordination

Revit 2026 MEP coordination, MVHR/ERV layout, airflow schedule, SolidWorks, AutoCAD/DXF, NZBC G4/H1 context, commissioning notes

- Built a NZ residential MVHR/ERV package with Revit MEP, CAD evidence and review notes.
- Modelled MVHR unit, ducts, terminals and hangers in SolidWorks; exported STEP, STL, DXF and PNG.
- Created a Revit 2026 model with plan, 3D view, airflow schedule, sheets and simple clash notes.
- Checked 60 L/s supply, 45 L/s extract and 85 L/s boost against duct velocity and pressure allowances.
- Added M-102 pressure review, BIM issue log and Mitsubishi Lossnay LGH-35RVX3-E selection note.
- Linked the design to NZBC G4/H1, Healthy Homes context and a commissioning test plan.

Manufacturing Workshop Training & DXF Preparation

Laser cutting, AutoCAD/DXF, CNC machining exposure, 3D printing workflow

- Prepared AutoCAD/DXF geometry for laser cutting and checked the cut plate against drawing intent.
- Practised CNC workflow: workholding, tool motion, cutting fluid, chip formation and shop safety.
- Reviewed machined gears and shafts for tolerance, edge quality and tool-access constraints.

Composite Formula SAE Front Wing Analysis

Abaqus/Standard, shell FEA, composite layup, structural checks

- Built an Abaqus shell FEA model with S4R elements, laminate layups and local mesh refinement.
- Converted aero, lateral cornering and 4.1 G bump conditions into structural load cases.
- Checked ply-level stress use; peak utilisation was about 74% under the 4.1 G bump case.

TECHNICAL SKILLS

- Proficient in: SolidWorks assemblies, enclosure features, gasket/screw-boss layout, STEP/STL export, component documentation
- Familiar with: STM32 C, Keil 5, Proteus, HX711, sensors, battery-monitor logic, serial/Bluetooth communication, PCB/circuit basics
- Engineering methods: DFM notes, FEA-style screening, test-list writing, firmware state-machine documentation, source-linked portfolio evidence